

CA2352: Unsafe DataSet or DataTable in serializable type can be vulnerable to remote code execution attacks (code analysis)

CA2352: Unsafe DataSet or DataTable in serializable type can be vulnerable to remote code execution attacks (code analysis) - dotpaul, learn.microsoft.com.

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Content

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CA2352: Unsafe DataSet or DataTable in serializable type can be vulnerable to remote code execution attacks

Summarize this article for me

Property	Value	Rule ID	CA2352	Title	Unsafe DataSet or DataTable in serializable type can be vulnerable to remote code execution attacks	Category	Security	Fix is breaking or non-breaking	Non-breaking	Enabled by default in .NET 10	No	Applicable languages	C# and Visual Basic
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Cause

A class or struct marked with `SerializableAttribute` contains a `DataSet` or `DataTable` field or property, and doesn't have a `DesignerCategoryAttribute`.

[CA2362](#) is a similar rule, for when there is a `DesignerCategoryAttribute`.

Rule description

When deserializing untrusted input with [BinaryFormatter](#) and the deserialized object graph contains a [DataSet](#) or [DataTable](#), an attacker can craft a malicious payload to perform a remote code execution attack.

This rule finds types which are insecure when deserialized. If your code doesn't deserialize the types found, then you don't have a deserialization vulnerability.

For more information, see [DataSet and DataTable security guidance](#).

How to fix violations

- If possible, use [Entity Framework](#) rather than [DataSet](#) and [DataTable](#).
- Make the serialized data tamper-proof. After serialization, cryptographically sign the serialized data. Before deserialization, validate the cryptographic signature. Protect the cryptographic key from being disclosed and design for key rotations.

When to suppress warnings

It's safe to suppress a warning from this rule if:

- The type found by this rule is never deserialized, either directly or indirectly.
- You know the input is trusted. Consider that your application's trust boundary and data flows may change over time.
- You've taken one of the precautions in [How to fix violations]().

Suppress a warning

If you just want to suppress a single violation, add preprocessor directives to your source file to disable and then re-enable the rule.

```
#pragma warning disable CA2352
// The code that's violating the rule is on this line.
#pragma warning restore CA2352
```

To disable the rule for a file, folder, or project, set its severity to `none` in the [configuration file](#).

```
[*.{cs,vb}]
dotnet_diagnostic.CA2352.severity = none
```

For more information, see [How to suppress code analysis warnings](#).

Pseudo-code examples

Violation

```
using System.Data;
using System.Runtime.Serialization;

[Serializable]
public class MyClass
{
    public DataSet MyDataSet { get; set; }
}
```

CA2350: Ensure DataTable.ReadXml()'s input is trusted

CA2351: Ensure DataSet.ReadXml()'s input is trusted

CA2353: Unsafe DataSet or DataTable in serializable type

CA2354: Unsafe DataSet or DataTable in deserialized object graph can be vulnerable to remote code execution attack

CA2355: Unsafe DataSet or DataTable in deserialized object graph

CA2356: Unsafe DataSet or DataTable in web deserialized object graph

CA2361: Ensure autogenerated class containing DataSet.ReadXml() is not used with untrusted data

CA2362: Unsafe DataSet or DataTable in autogenerated serializable type can be vulnerable to remote code execution attacks